

Solids Handling

PDF

Hopper Design Principles

January 1, 2016 | By Greg Mehos and Dave Morgan, Jenike & Johanson

When hoppers are designed without consideration of the actual materials being handled, problems inevitably arise. Follow this guidance to avoid common solids-handling issues, such as erratic flow and no flow

Pivotal work on the development of the theory of bulk solids flow began in earnest in the early 1950s, when Andrew Jenike applied a solids-mechanics-continuum concept to develop a logical, theoretical approach for understanding and managing solids flow. He developed testing methods, equipment, and design techniques and conducted experiments to confirm and refine his groundbreaking analysis [1].

Prior to Jenike’s work, bins and hoppers were typically designed primarily from an architectural or fabrication standpoint (for instance, hopper walls were sloped 30 deg from vertical to reduce the waste of wall materials, or 45 deg to minimize headroom requirements and simplify design calculations). However, extensive experience has shown that designing equipment without regard to the actual bulk materials being handled often leads to flow problems, such as arching, ratholing, erratic flow and even no flow. By measuring the flow properties of a bulk solid, the flow behavior of the material can be predicted, and more reliable hoppers and bins can then be designed.

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Values of the allowable hopper angle θ' (measured from vertical) are on the x-axis, and values of the wall friction angle ϕ' are on the y-axis. Any combination of ϕ' and θ' that falls within the limiting mass flow region of the chart will provide mass flow.

Hoppers with round or square outlets should not be designed at the theoretical mass-flow hopper-angle value. Otherwise, a small change in powder properties may cause the flow pattern inside the hopper to change from mass flow to funnel flow — bringing its associated risk of flow problems. A 3-deg margin of safety (with respect to the mass-flow hopper angle given in Figure 4) is recommended.

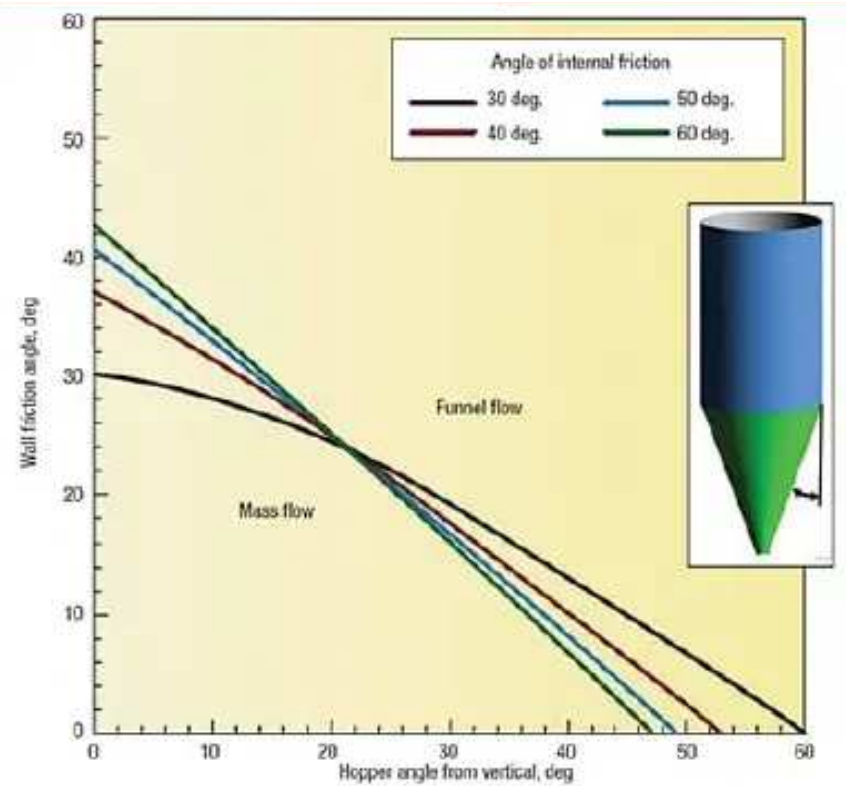


FIGURE 4. This plot shows the theoretical mass-flow hopper angles for hoppers with round or square outlets. Note: a minimum safety factor of 3 deg should be used

Sloping walls required for mass flow in wedge-shaped hoppers can be 10–12-deg less steep than those required to ensure mass flow in conical hoppers. Wedge-shaped and transition hoppers are therefore frequently used for materials that have high wall friction.

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NOMENCLATURE

B	Outlet diameter or width, m or ft
B_{min}	Minimum outlet diameter or width to prevent arching, m or ft
D_f	Critical rathole diameter, m or ft
dP/dz	Pressure gradient, Pa/m or lb/ft ³
FF	Flow Function
f_c	Cohesive strength
ff	Flow factor
$G(\phi)$	Function defined by Jenike [1]; as shown in Figure 11
g	Acceleration due to gravity, m/s ² or ft/s ²
$H(\Theta')$	Function defined by Jenike [1]; as shown in Figure 11
h	Bed depth, m or ft
k	Janssen coefficient
m	Parameter equal to 0 for slotted outlets, and equal to 1 for round outlets
R_H	Hydraulic radius, m or ft
u_o	Velocity of solids at outlet, m/s or ft/s
δ	Effective angle of internal friction, deg
ϕ	Static angle of internal friction, deg
ϕ'	Wall friction angle, deg
ρ_b	Bulk density, kg/m ³ or lb/ft ³
ρ_{bo}	Bulk density at outlet, kg/m ³ or lb/ft ³
σ_1	Major consolidation stress, Pa or lb/ft ²
σ_{crit}	Critical stress, Pa or lb/ft ²
σ_{ss}	Steady-state normal stress, Pa or lb/ft ²
$\bar{\sigma}$	Stress on the abutment of an arch, Pa or lb/ft ²
Θ'	Hopper angle from vertical, deg
τ_{ss}	Steady-state shear stress, Pa or lb/ft ²

Minimum outlet dimension

The outlet of the hopper section must be large enough to prevent cohesive arches or stable ratholes from developing. The required outlet size depends on the cohesive strength and the bulk density of the powder. The cohesive strength is measured by shear-cell testing, as described in ASTM D-1628 [2] and D-6773 [3].

Figure 6 shows schematic diagrams for two common shear-cell testers. A sample of powder is placed in a cell and then pre-sheared – that is, the sample is consolidated by exerting a normal stress, and then sheared until the measured shear stress is steady. (This is

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Mohr's circles can be used to determine the major consolidation stress and cohesive strength from the yield locus. The major consolidation stress is the value of the intersection of the horizontal axis and a Mohr's circle, drawn through the steady-state point and tangent to the yield locus. The cohesive strength is related to the state of stress where a free boundary exists. Hence, the cohesive strength is the intersection of the horizontal axis and a Mohr's circle that passes through the origin (where stress equals zero) and is tangent to the yield locus. The kinematic angle of internal friction (ϕ) and effective angle of internal friction (δ) can also be determined, as shown in Figure 7.

By conducting the test over a range of consolidation states, the relationship between consolidation pressure and the cohesive strength of the bulk material can be established, following a procedure established and described by Jenike [1]. This relationship is commonly called the material's Flow Function. An example of a powder's Flow Function is given in Figure 8.

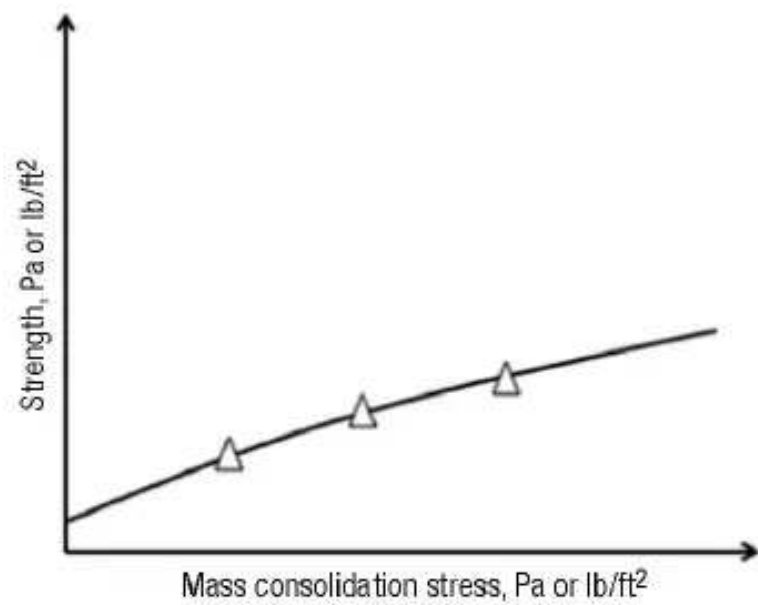


FIGURE 8. The relationship between the major consolidation stress and the cohesive strength is called the Flow Function



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$$ff = \frac{\sigma_1}{\bar{\sigma}_1} \quad (1)$$

The flow factor is a function of the powder’s effective angle of internal friction, the hopper angle, and the wall friction angle. Charts that provide flow factors for conical and wedge-shaped hoppers are given in Jenike [1]; Examples are shown in Figures 9a and 9b. Typical values of the flow factor range between 1.1 and 1.7.

Superimposing the material’s Flow Function and flow factor on the same graph allows the cohesive strength and arch stress to be compared. The flow factor is constructed by drawing a line having a slope equal to 1/*ff* through the origin.

There are three possibilities:

- *The Flow Function lies below the flow factor, and the two curves do not intersect.* When this is the case, the stress imparted on the arch is always greater than the material’s cohesive strength, and there is no minimum outlet dimension requirement to prevent cohesive arching
- *The Flow Function lies above the flow factor and the curves do not intersect.* The bulk solid will not flow due to gravity alone, and another means of discharging the powder must be employed
- *The Flow Function and flow factor intersect, as shown in Figure 10.* At the point where the two lines intersect, the arch stress and the cohesive strength of the bulk solid are the same and equal to the critical stress (σ_{crit})



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If a hopper with a circular outlet is designed with an opening large enough to prevent the development of a stable rathole, cohesive arching will not occur. For wedge-shaped hoppers, the width of the slotted outlet must be large enough to prevent a cohesive arch from developing. The same procedure that is used to determine the minimum outlet width to prevent arching in a wedge-shaped mass–flow hopper is followed, except that a flow factor of 1.7 is used.

Note that these analyses assume continuous handling of the powder. If the powder is to be stored at rest for a period of time, time tests should be conducted. Time tests are described in ASTM D-6128 and D-6773 [2, 3].

Feeder considerations

Feeders can also be a source of hopper flow problems if the incorrect devices are used or if they are improperly designed. This is especially true for hoppers with slotted outlets, where feeders should be designed to draw uniformly along the entire cross-section of the outlet in order for mass flow to occur. However, even hoppers with round outlets can have uneven flow if a proper interface is not utilized.

Rotary valves are often used beneath hoppers with round outlets. They are particularly useful for applications where a seal must be provided to prevent air from flowing out of or into the hopper outlet. If a rotary valve is used, a short vertical spool section should be installed between the hopper outlet and valve inlet. Otherwise, material may flow preferentially from the upside of the valve and affect the flow pattern inside the vessel. This is shown in Figure 12.



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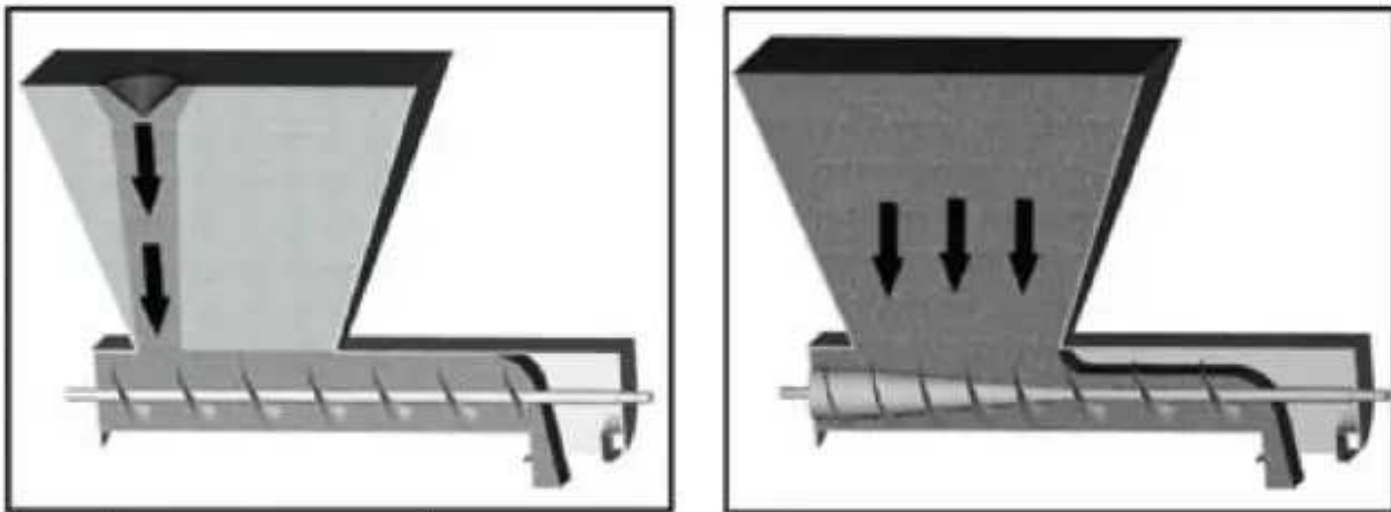


FIGURE 13. A screw feeder with a constant diameter and a constant pitch screw results in funnel flow. A mass-flow screw feeder where the screw has a tapered shaft and increasing pitch sections ensures that all material will flow when discharged

A mass flow screw feeder, comprised of a tapered section followed by a section with increasing pitch, ensures that the capacity of the feeder increases in the direction of flow (see Figure 13). The length of the cone and the pitch schedule are chosen such that the capacity of the screw increases linearly along the hopper length. The length of the screw must be between three and six times the width of the hopper to meet fabrication tolerances, and the screw diameter is equal to the hopper width.

Discharge rate

For coarse powders, the maximum discharge rate from a mass flow hopper can be calculated using Equation (5):

$$u_o = \sqrt{\frac{Bg}{2(1+m)\tan(\theta')}} \tag{5}$$

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The magnitude of the pressure gradient depends on the bulk density and the permeability of the powder. Note that because of the vacuum that develops above the hopper outlet, the pressure gradient, dP/dz , is negative. See Schulze [4] for a discussion of the challenges of accurately predicting the pressure gradient.

Final remarks

Compared to liquids and gases, chemical engineers’ training in bulk-solids handling is often lacking. By understanding the fundamental principles and conducting proper flow properties tests, a chemical engineer can design reliable systems for storing and handling bulk solids. Knowing the relationship between a material’s cohesive strength and consolidation stress allows the engineer to calculate hopper-outlet dimensions that will prevent flow obstructions from developing. Similarly, wall-friction test results allow engineers to determine hopper geometries that are required to promote mass flow. Permeability tests are required to calculate limiting mass-flow discharge rates.

Edited by Suzanne Shelley

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2. ASTM D-6128, Standard Test Method for Shear Testing of Bulk Solids Using the Jenike Shear Cell, ASTM International, 2006.
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